Econometrics Part 5

Endogeneity

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Outline

1. Framework
2. Non-exogeneity
3. The IV estimator
4. IV in practice

Considering random explanatory variables

- In the basic framework, explanatory variables X are usually considered not random
- This amounts to reasoning conditionally on X, i.e. as if the X were fixed
- An important hypothesis was that X and u were independent, which is always the case if X is fixed
- But the X's can be considered as random variables
- All the previous results still hold (as long as $E(u|X) = 0$), because they were obtained conditionally on X

Assumptions for estimation

1. $E(u) = 0$
2. X is random and $E(u|X) = 0$: in particular, $\forall t, t'$, X_t is uncorrelated with $u_{t'}$
3. $\text{rank}(X) = k$
4. $E(uu') = \sigma^2 I_N$
5. Error terms are iid $(0, \sigma^2)$
6. $\text{plim}(X'X/N) = V_X$, which is a positive definite matrix (when N goes to infinity, the X variables always keep some variance)

Properties of OLS: unbiasedness

- $\hat{b}_{ols}$ is still unbiased
- We know that $\hat{b}_{ols} = (X'X)^{-1}X'y = b + (X'X)^{-1}X'u$
- So $E(\hat{b}_{ols}) = b + E[(X'X)^{-1}X'u]$
- We get: $E[(X'X)^{-1}X'u] = E_X\left[E_u[(X'X)^{-1}X'u \mid X]\right]$
  $= E_X\left[(X'X)^{-1}X'E_u[u \mid X]\right]$ and $E_u[u \mid X] = 0$
- In the same way, we get an expression for the variance of $\hat{b}_{ols}$:
- $V(\hat{b}_{ols}) = \sigma^2 E[(X'X)^{-1}]$

Properties of OLS: consistency

$\hat{b}_{ols}$ is still consistent:

$$\text{plim}(\hat{b}_{ols}) = b + \left[\text{plim}\left(\frac{X'X}{N}\right)\right]^{-1} \text{plim}\left(\frac{X'u}{N}\right) = b + V_X^{-1} \cdot 0 = b$$

We will see on the next slide that the asymptotic distribution of $\hat{b}$ is a normal, as expected.

Asymptotics: OLS estimators

We wish now to enlarge our framework to the case where u's are only iid (and not necessarily normal) and where X's can be random (and not necessarily fixed). In what follows, we will thus use the following asymptotic results, obtained under the usual model hypotheses:

$$\text{plim}(\hat{b}_{ols}) = b + \left[\text{plim}\left(\frac{X'X}{N}\right)\right]^{-1} \text{plim}\left(\frac{X'u}{N}\right) = b + V_X^{-1} \cdot 0 = b$$

$$\text{plim}(\hat{\sigma}^2_{ols}) = \sigma^2$$

$$\sqrt{N}(\hat{b}_{ols} - b) \xrightarrow{d} \mathcal{N}(0, \sigma^2 V_X^{-1})$$

Asymptotic tests

Tests of r linear constraints (C is of size (r, k)) are run using the following formula:

$$f = \frac{(C\hat{b}_{ols} - Cb)'(C(X'X)^{-1}C')^{-1}(C\hat{b}_{ols} - Cb)}{\sigma^2} \xrightarrow{d} \chi^2_r$$

Where, as before in the finite sample case, σ^2 is unknown. But since by assumption we work with a very large sample, $\hat{\sigma}^2_{ols}$ is a consistent estimate of σ^2. We can thus replace σ^2 by its consistent estimate, without having to replace the χ^2 with a Fisher.

General properties

If we assume that the X_i are iid and the u's are normal, then:

- OLS estimators are consistent, and normal conditionally on X
- The T- and F-statistics keep their Student and Fisher distributions (their conditional distribution does not depend on X): we can run the usual T- and F-tests
- If the u are not normal but only iid, we need to make sure the sample is large, and use the asymptotic version of the usual T- and F-tests
- In that case, Normal replaces Student and χ^2 replaces Fisher

Non-exogenous explanatory variables

- Let's use the following model: $y_t = X_t' b + u_t$
- Assume that $E(X_t u_t) \neq 0$: some explanatory variables are correlated with the current error term
- In this case, $\text{plim}(X'u/N) \neq 0$
- OLS is biased and inconsistent

$$\text{plim}(\hat{b}_{ols}) = b + \left[\text{plim}\left(\frac{X'X}{N}\right)\right]^{-1} \text{plim}\left(\frac{X'u}{N}\right) \neq b$$

Even if we increase the size of the sample, we won't get the right value for b.

Exogeneity: definitions

- **Case 1**: Strict exogeneity: $\forall t, t', E(X_t u_{t'}) = 0$
- X and u are never correlated
- **Case 2**: (Contemporaneous) exogeneity: $\forall t, E(X_t u_t) = 0$
- X and u are uncorrelated for any given time period
- **Case 3**: X predetermined: $\forall t' \geq t, E(X_t u_{t'}) = 0$
- X uncorrelated with current and future u

This class focuses on cross-sections, so on the second case.

Omitted variables

Let's assume the true model is the following:

$$y_t = X_t'b + w_t d + v_t$$

If we omit w_t and estimate the following model instead:

$$y_t = X_t'b + u_t$$

- w_t is included in u_t
- If w_t is correlated with X, then $E(X_t w_t) \neq 0$ and thus $E(X_t u_t) \neq 0$: OLS is biased
- Remark: in the fixed X case, we would get the same result using the Frisch-Waugh theorem
- When a variable that is correlated with the X's is omitted, estimators suffer from an *omitted variable bias*
- Estimators are inconsistent

Variables measured with error

Let's use a very simple one-variable theoretical model (no constant if variables are centered):

$$y_t^* = x_t^* b + v_t$$

Assume that x is measured with error. To run the estimation, we have access to values $y_t = y_t^*$ and $x_t = x_t^* + e_t$, e_t being a white noise. The estimated model is thus $y_t = x_t b + u_t$, with $u_t = v_t - be_t$. We thus have:

$$E(x_t u_t) = E[(x_t^* + e_t)(v_t - be_t)] = -b\sigma_e^2 \neq 0$$

OLS is thus inconsistent, and biased towards 0 (*attenuation bias*):
$\text{plim}(\hat{b}_{ols}) = b\,\sigma_{x^*}^2/(\sigma_{x^*}^2 + \sigma_e^2)$, so the influence of x on y is underestimated.

Simultaneous equations

Say we have the following system of equations:

- $Y_t = a + bX_t + u_t$ (1)
- $X_t = Y_t + Z_t$ (2)
- Because of equation (1), Y_t is endogenous, and because of equation (2), X_t is endogenous too
- At the system level, only Z_t is exogenous
- To get back to what we are used to, we should rewrite endogenous variables as functions of exogenous variables only

The system can be rewritten (if $b \neq 1$): $Y_t = \frac{a}{1-b} + \frac{b}{1-b}Z_t + \frac{1}{1-b}u_t$ and $X_t = \frac{a}{1-b} + \frac{1}{1-b}Z_t + \frac{1}{1-b}u_t$. Notice that X_t is a function of u_t: in equation (1), it is thus correlated with the error term. A basic hypothesis of OLS is violated, and if we estimate (1) without taking into account the information provided by (2), estimates will be inconsistent.

Autocorrelation with lagged dependent variable

Assume we use the following model:

$$y_t = b_0 + b_1 x_t + b_2 y_{t-1} + u_t$$

With: $u_t = \rho u_{t-1} + \varepsilon_t$
We can then write:

- $y_t = b_0 + b_1 x_t + b_2 y_{t-1} + \rho u_{t-1} + \varepsilon_t$ (1)
- And at the same time: $y_{t-1} = b_0 + b_1 x_{t-1} + b_2 y_{t-2} + u_{t-1}$ (2)
- Equation (2) shows that y_{t-1} depends on u_{t-1}, so y_{t-1} is correlated with the error term $\rho u_{t-1} + \varepsilon_t$ in equation (1)
- The lagged outcome variable is thus not exogenous in the estimated model
- If error terms are not autocorrelated ($\rho = 0$), y_{t-1} is uncorrelated with u_t: it is predetermined (case 3)
- How to test for this: Durbin's h statistic

General issue in policy evaluation

- Say we want to evaluate the impact of a policy on people's wages (e.g. a training program)
- A model describing the wage outcome is $Y_i = a + bX_i + cP_i + u_i$, where X comprises individual characteristics and P is a dummy variable indicating whether the individual was assigned to the program
- If the policy is not *randomized*, i.e. if being assigned to the program depends on unobserved individual characteristics, then P is correlated with u and the impact of the policy cannot be estimated consistently
- Intuitive reason: those who were assigned to the program are not comparable to those who were not, so the latter cannot be considered as a valid control group for the former
- This is called a *selection effect*

Consequences

In the case of non-exogeneity (also called endogeneity) of some explanatory variables:

- OLS estimators are biased: on average, we do not get the true value of parameters
- OLS estimators are inconsistent: even if we increase the size of the sample, the bias does not go to zero and we will never get the true value of parameters

We thus have to use another estimation technique, using auxiliary variables: the instrumental variables technique.

Instrumental variables

Let's use the following model, where (to make things simple) *all* X's are endogenous, except the constant (which can't be endogenous by definition):

$$y = Xb + u$$

Let's consider a set of variables $Z_{(N,p)}$ different from the X's but including the constant, with the following properties:

- $E(Z_t u_t) = 0$: variables Z are exogenous
- $\text{rank}(Z) = p$
- $\text{plim}(Z'X/N) = V_{ZX}$, with V_{ZX} a non-null matrix of dimension (p, k) and rank k
- $\text{plim}(Z'Z/N) = V_Z$, with V_Z a finite positive definite matrix of dimension (p, p)

Meaning: variables Z need to be both exogenous and correlated with X, and the number of instruments p must be at least the number of explanatory variables k ($p \geq k$).

The estimator

The original model is:

$$y = Xb + u$$

- Assume we regress X on variables Z
- We compute their predictions $\tilde{X}$
- We use these new variables in the model instead:

$$y = \tilde{X}b + u$$

- It should work because $\tilde{X}$ is only made up of Z that are exogenous: $\tilde{X}$ is thus exogenous too

Intuition (1)

- OLS on the original model $y = Xb + u$ are inconsistent because variables X are correlated with u
- We can't replace the X's by other variables because the model should remain unchanged
- We would like to get rid of this correlation, while keeping the information provided by X
- Imagine a regression where each variable in X would be explained by the set of variables Z, say: $\forall j,\ x_j = Zd + e$
- A prediction for x_j would be: $\hat{x}_j = Z\hat{d}$
- If this regression is good enough, $\hat{x}_j$ is close to the true x_j
- And $\hat{x}_j$ is a linear combination of variables Z that are exogenous: it is thus exogenous as well
- We can thus replace x_j by $\hat{x}_j$ in the original model

Transforming the model

The original model is:

$$y = Xb + u$$

The new model is:

$$y = \tilde{X}b + u$$

- Problem: the new model is not directly equivalent to the original one: only X was modified
- How can we make the new model equivalent to the old one?
- Notice that $\tilde{X}$ is the prediction of X using the Z
- So that $\tilde{X} = P_Z X$
- We need to multiply every term of the original model by P_Z

The IV estimator

- Let's call P_Z the orthogonal projection matrix that projects on $L(Z)$: $P_Z = Z(Z'Z)^{-1}Z'$
- P_Z is symmetric ($P'_Z = P_Z$) and idempotent ($P_Z P_Z = P_Z$)
- Let's define $\tilde{y} = P_Z y$, $\tilde{u} = P_Z u$ and $\tilde{X} = P_Z X$
- It's as if we had premultiplied the original model by P_Z: $\tilde{y} = \tilde{X}b + \tilde{u}$
- The IV estimator is: $\hat{b}_{iv} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{y} = (X'P_Z X)^{-1}X'P_Z y$

Remark

- To get to the estimated model, we multiply both y and X on the left by P_Z
- What would happen if we multiplied only X and not y?
- $(\tilde{X}'\tilde{X})^{-1}\tilde{X}'y = (X'P_ZX)^{-1}X'P_Zy = \hat{b}_{iv}$
- It does not change anything as regards $\hat{b}_{iv}$: in practice, we transform only X and leave y unchanged

Intuition (2)

We can also explain this geometrically:

- To remove the correlation between X and u, we project the model onto the vector space $L(Z)$, spanned by the columns of Z, which are at the same time exogenous and correlated with X
- This amounts to keeping from X only the exogenous information, uncorrelated with the error terms
- The more the Z are correlated with the X (and the more numerous the Z's are), the more precise the estimator is, because the information loss is minimal
- $P_Z y$ is the prediction from the regression of y on variables Z, same for $P_Z X$

Remarks

- If $p = k$ (same number of instruments Z and of explanatory variables X), we get $\hat{b}_{iv} = (Z'X)^{-1}Z'y$
- Proof: rewriting $\hat{b}_{iv}$, given that in this case, matrix $Z'X$ is square and invertible
- But with this minimal number of instruments, we cannot test the validity of the instruments (Sargan test, see below)

Generalization (1)

- What if some variables in X are endogenous, and others exogenous?
- Let's call X_1 the set of the exogenous ones and X_2 the set of the endogenous ones: the matrix notation would be $X = (X_1, X_2)$ (again, the constant is always exogenous)
- In that case, variables X_1 can be used as instruments in addition to the Z
- Let's call W this extended set of instruments: $W = (Z, X_1)$
- To compute the IV estimator, we premultiply X by P_W
- $P_W X = P_W(X_1, X_2) = (P_W X_1, P_W X_2) = (X_1, P_W X_2)$
- Indeed, since X_1 belongs to W, it is projected onto itself and remains unchanged

Generalization (2)

- Conclusion: we will use X_1 as additional instruments because they will remain unchanged in the transformed model, and will help to instrument the X_2
- We only need to make sure that there are at least as many "new" instruments Z as endogenous variables X_2, and don't rely only on the exogenous X_1
- Z are sometimes called *excluded instruments*
- We realize here that the constant, exogenous by nature, belongs to the exogenous X_1, and that we were right to use it as an instrument in the first place
- That's what we did when describing the first stage regression, which contains a constant by default

Important: final notation

- Since the exogenous variables from the set of variables X will always be used as instruments, we don't usually bother mentioning it
- **We usually call Z the global matrix of instruments that contains both the excluded instruments and the variables from X that are exogenous**
- So that the hypotheses and formulae are the same as the ones presented in the beginning
- We only need to make sure that there are at least as many excluded instruments as endogenous variables

Properties of the IV estimator (1)

- $\hat{b}_{iv}$ is biased in a small sample
- We cannot derive a general expression for its finite-sample variance-covariance matrix
- But it is consistent (when the sample size goes to infinity)
- It is asymptotically normal:

$$\sqrt{N}(\hat{b}_{iv} - b) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2 (V'_{ZX} V_Z^{-1} V_{ZX})^{-1}\right)$$

Properties of the IV estimator (2)

There is no general expression for its variance-covariance matrix, but we can derive its *asymptotic* variance-covariance matrix, which is consistently estimated by:

$$\widehat{V}_{asympt}\left(\sqrt{N}(\hat{b}_{iv} - b)\right) = \hat{\sigma}^2_{iv}\left(\frac{X'P_Z X}{N}\right)^{-1}$$

with $\hat{\sigma}^2_{iv} = \frac{SSR_{iv}}{N} = \frac{\hat{u}'\hat{u}}{N}$ and $\hat{u} = y - X\hat{b}_{iv}$

- When OLS is consistent, IV is less precise than OLS
- Precision increases with the number of instruments (asymptotically)

Tests on regression parameters

- We do not have convenient properties of the estimator in the finite sample case
- We can only run asymptotic tests: Wald tests, similar to F-tests, but when the sample size is large
- Example of a test for r linear constraints on parameters:

$$f = \frac{(C\hat{b}_{iv} - Cb)'[C(X'P_Z X)^{-1}C']^{-1}(C\hat{b}_{iv} - Cb)}{\hat{\sigma}^2_{iv}} \xrightarrow{d} \chi^2_r$$

With r the number of linear restrictions: C is of size (r, k).
Proof: taking the expression of the asymptotic normality of $\hat{b}_{iv}$, taking its quadratic form and "dividing" it by its variance gives a χ^2 distribution (the "true" variance is replaced by its consistent estimator).

How to run IV estimation

- Python: `IV2SLS` from `statsmodels.sandbox.regression.gmm` (basic), or the `linearmodels` package (`linearmodels.iv.IV2SLS`), which also reports first-stage statistics and the exogeneity and Sargan tests
- This amounts to running two-stage least squares
- Intuition: first, regress the y and X's on variables Z, then use the predictions in the model instead of the original values

Two-stage least squares (1)

- Run $k+1$ regressions, to get $P_Z y$ and $P_Z X$
- Estimate OLS on the transformed model $P_Z y = P_Z X b + \tilde{u}$
- We thus get $\hat{b}_{iv} = (X' P_Z X)^{-1} X' P_Z y$
- The first-stage regressions can be used to assess the relevance of the instruments (they have to be correlated enough with the X's)
- Remark: this provides the same values if we do not replace y by $P_Z y$

Two-stage least squares (2)

Warning: if we do this procedure "by hand", running 2 OLS regressions, instead of running the dedicated IV procedure of the software, the standard errors of the second regression cannot be used for tests on the coefficients

Reason: in the second stage equation, residuals are computed as: $\hat{u} = P_Z y - P_Z X \hat{b}_{iv}$ (or $y - P_Z X \hat{b}_{iv}$)

Whereas they should be computed as $\hat{u} = y - X \hat{b}_{iv}$

Exogeneity test (1)

- We test H_0: $E(X'u) = 0$
- This is called the "Hausman test" or "Durbin-Wu-Hausman test"
- If H_0 is true, then both OLS and IV estimators are consistent (and OLS is more precise)
- If H_0 is false, only the IV estimator is consistent
- The test is based on the difference between $\hat{b}_{iv}$ and $\hat{b}_{ols}$
- They are asymptotically normal: if we compute the difference between the two, take its quadratic form and "divide" it by its variance matrix, we will get a χ^2 distribution

Exogeneity test (2)

One could think that the parameter of this χ^2 is the number of tested variables (those potentially endogenous), just like a Wald test. But looking closely at the test statistic:

$$H = (\hat{b}_{iv} - \hat{b}_{ols})'(V(\hat{b}_{iv}) - V(\hat{b}_{ols}))^{-1}(\hat{b}_{iv} - \hat{b}_{ols})$$

This parameter is in fact equal to the rank of the matrix $V(\hat{b}_{iv}) - V(\hat{b}_{ols})$ (whose inverse is then a generalized inverse).
In finite samples, the estimated difference of variances may not be positive definite, so sometimes the software can't run the test (or finds $H < 0$).

The augmented regression

- Consider model $y = Xb + u$, where a subset x of the variables belonging to X might be endogenous
- Let's call Z the instruments, some belonging to X (in fact the X without the x) and some not
- Consider the augmented model: $y = Xb + M_Z xc + \varepsilon$
- $M_Z x$ are the residuals of the regression of x on Z
- The $\hat{b}$ of this "augmented" regression is equal to the IV estimator of the original model
- Testing $c = 0$ amounts to testing the exogeneity of x (it is equivalent to the Hausman test)

Proofs

- $\hat{b}_{aug} = \hat{b}_{iv}$ and equivalence of tests: using the Frisch-Waugh theorem
- Remark: this augmented model has no theoretical meaning, it's only a tool

Selecting valid instruments

Sargan test: H_0: $E(Z'u) = 0$
Also called: test of overidentifying restrictions
Under H_0:

$$\frac{\hat{u}' P_Z \hat{u}}{s^2} \xrightarrow{d} \chi^2_{p-k}$$

with $\hat{u} = y - X\hat{b}_{iv}$ and $s^2 = \frac{\hat{u}'\hat{u}}{N}$.
$\hat{u}' P_Z \hat{u}$ is the sum of the squared fitted values of the regression of $\hat{u}$ on Z (the statistic is NR^2 of this regression).
Remark: when $p = k$, the statistic is always zero because $\hat{b}_{iv} = (Z'X)^{-1}Z'y$ and $Z'\hat{u} = 0$. So we cannot run the test with the minimum number of instruments.

The problem with weak instruments

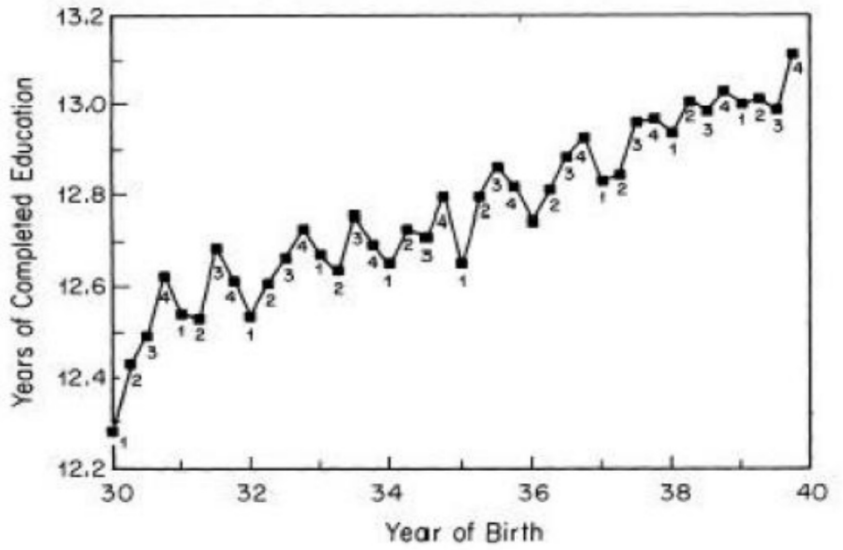
- If instruments are too weakly correlated with the X's, even if we increase the number of observations, there can be an important bias in estimations
- Plus, the estimator has a non-normal sampling distribution, which makes statistical inference meaningless
- The weak instrument problem is increased with many instruments, so drop the weakest ones and use the most relevant ones
- To measure how instruments are correlated with the potentially endogenous variables, run the first-stage regression explaining the latter by the instruments
- Criterion: the F statistic testing that the *excluded* instruments are jointly zero in the first stage (not the global F): if $F < 10$, instruments are weak (Staiger-Stock rule of thumb)

Returns to education

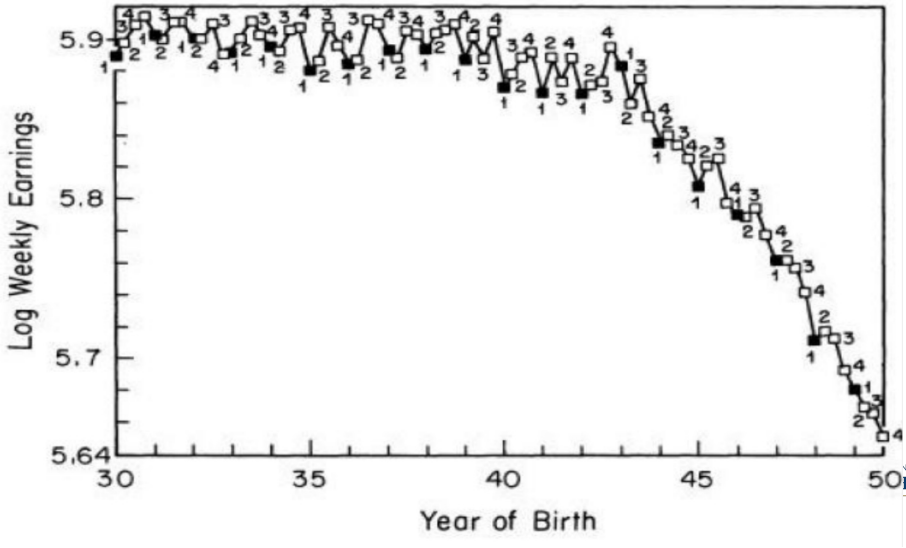
Angrist-Krueger (1991), "Does Compulsory School Attendance Affect Schooling and Earnings?", Quarterly Journal of Economics

- Goal: find the causal link between education and wages, knowing that such a wage equation is subject to endogeneity
- In the US, compulsory education starts the year pupils turn 6, and ends when they turn 16
- Pupils born in January begin school at 6 years 8 months old; those born in December, at 5 years 9 months old
- For pupils who stop school at 16, those born in December study almost one year more than those born in January
- Quarter of birth is thus correlated with schooling, but as good as random, so uncorrelated with individual ability: good instrument for schooling

Quarter of birth and education



Quarter of birth and earnings



Results

Independent variable	(1) OLS	(2) TSLS	(3) OLS	(4) TSLS
Years of education	0.0711	0.0891	0.0711	0.0760
	(0.0003)	(0.0161)	(0.0003)	(0.0290)
Race (1 = black)	—	—	—	—
SMSA (1 = center city)	—	—	—	—
Married (1 = married)	—	—	—	—
9 Year-of-birth dummies	Yes	Yes	Yes	Yes
8 Region-of-residence dummies	No	No	No	No
Age	—	—	−0.0772	−0.0801
			(0.0621)	(0.0645)
Age-squared	—	—	0.0008	0.0008
			(0.0007)	(0.0007)
χ^2 [dof]	—	25.4 [29]	—	23.1 [27]

Method

- Instruments: quarter-of-birth dummies, interacted with year-of-birth (and state-of-birth) dummies
- Result: OLS underestimates the impact of education on earnings
- Caveat: Bound, Jaeger and Baker (1995) showed that with so many weak instruments, 2SLS is biased towards OLS (see weak instruments)
- Even better method: randomized trials
- E.g. Moving to Opportunity, RAND Health Insurance Experiment
- Many other methods in policy evaluation: differences-in-differences, matching, etc.

Are IV estimates reliable?

- The goal of IV estimation is to bring to light real causality and not mere correlation between outcome y and explanatory variables X
- In treatment evaluation, we want to assess the Average Treatment Effect
- So we instrument the dummy indicating whether the individual received the treatment, because we suspect that taking the treatment is not exogenous: it is likely to be correlated with unobserved heterogeneity
- However, it can be shown that in doing so, we estimate a Local Average Treatment Effect (Imbens and Angrist, 1994), i.e. the impact of the treatment only on the individuals for whom the instrument indeed has an impact (called *compliers*)